3. SEQUENCE OPERATIONS AND SIGNAL ANALYSIS

Objectives:

This experiment is divided into four simulations:

1. Addition and Multiplication of Two Sequences

2. Energy and Power Calculation of Given Signals

3. Autocorrelation

4. Cross Correlation

Experiment 3.1

Addition and Multiplication of Two Sequences

Program:

```
% Addition and Multiplication of Two Sequences

clc;

clear;

close all;

% Sample index

n = 0:10;

% Sequence 1

x1 = [1 2 3 4 5 6 7 8 9 10 11];

% Sequence 2

x2 = [11 10 9 8 7 6 5 4 3 2 1];

% Addition of sequences
```

```
y_add = x1 + x2;

% Multiplication of sequences

y_mul = x1 .* x2;

% Plotting

figure;

subplot(2,2,1);

stem(n,x1,'filled');

grid on;

title('Sequence x_1(n)');

xlabel('n');

ylabel('Amplitude');

subplot(2,2,2);

stem(n,x2,'filled');

grid on;

title('Sequence x_2(n)');

xlabel('n');

ylabel('Amplitude');

subplot(2,2,3);

stem(n,y_add,'filled');

grid on;

title('Addition: x_1(n) + x_2(n)');

xlabel('n');
```

ylabel('Amplitude');

subplot(2,2,4);

stem(n,y_mul,'filled');

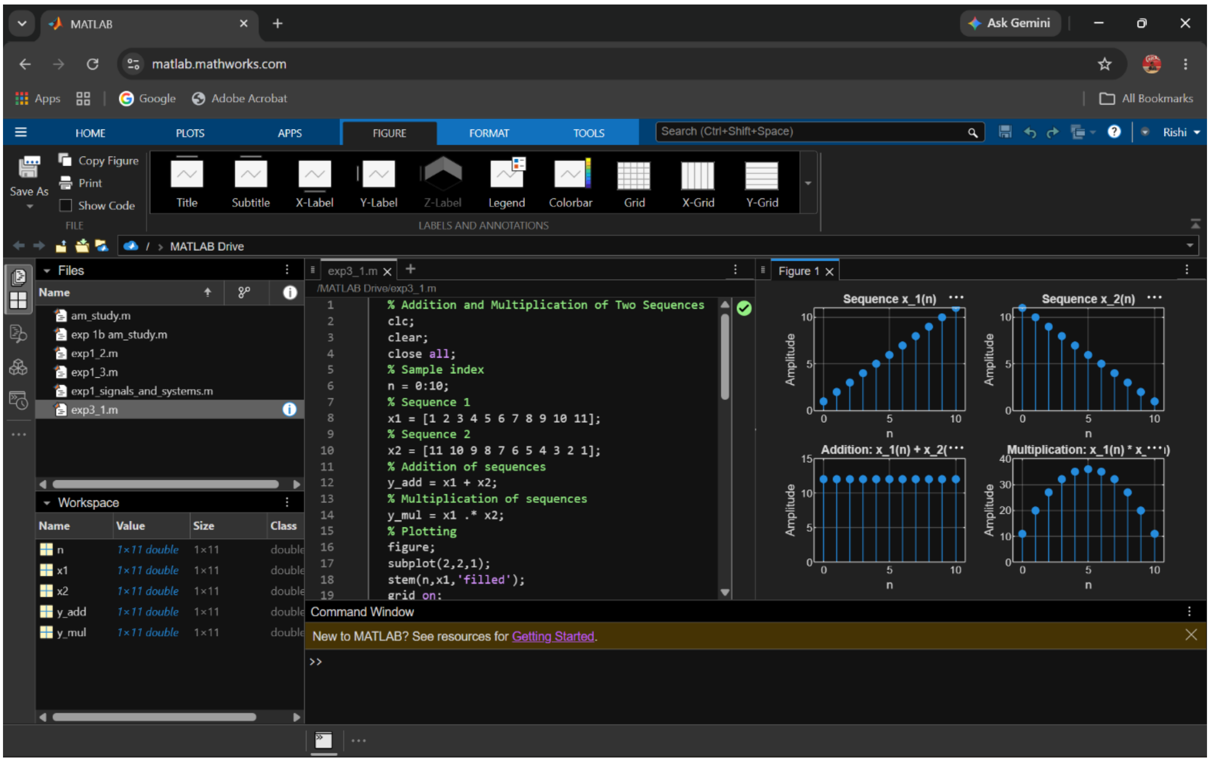
grid on;

title('Multiplication: x_1(n) * x_2(n)');

xlabel('n');

ylabel('Amplitude');

Output:



Experiment 3.2

Energy and Power Calculation of Given Signals

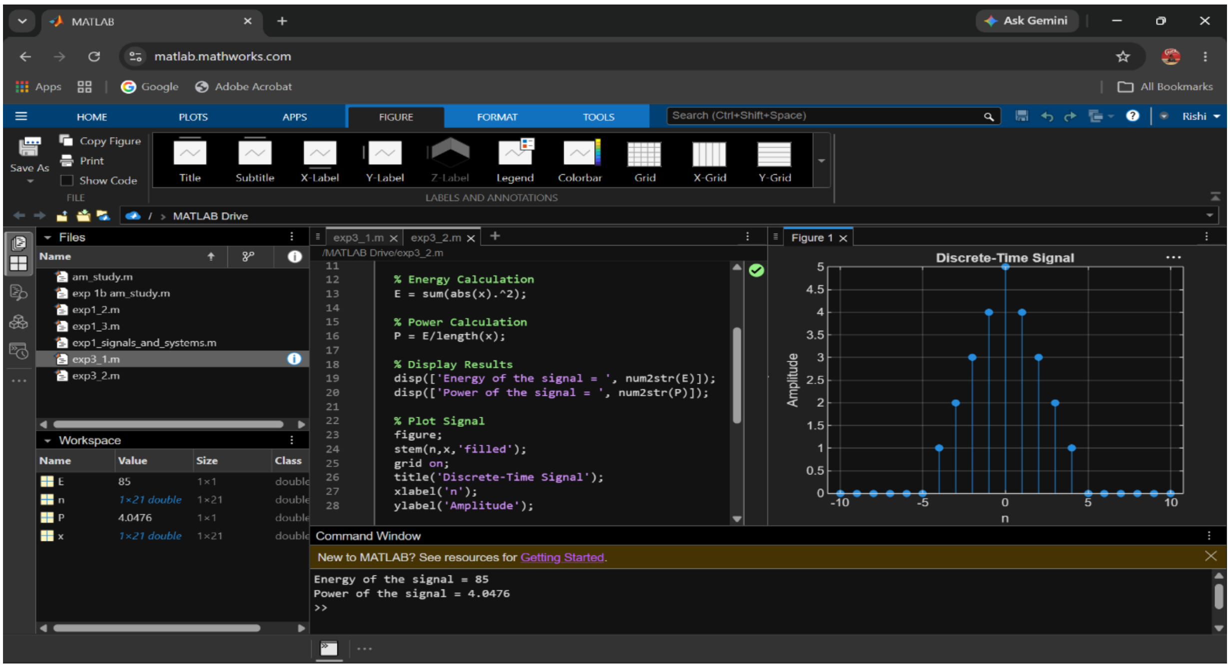
Program:

% Energy and Power Calculation of Signals

clc;

```
clear;

close all;

% Time index

n = -10:10;

% Signal

x = [zeros(1,10) 1 2 3 4 5 4 3 2 1 zeros(1,10)];

% Energy Calculation

E = sum(abs(x).^2);

% Power Calculation

P = E/length(x);

% Display Results

disp(['Energy of the signal = ', num2str(E)]);

disp(['Power of the signal = ', num2str(P)]);

% Plot Signal

figure;

stem(n,x,'filled');

grid on;

title('Discrete-Time Signal');

xlabel('n');

ylabel('Amplitude');
```

Output:



4. SYSTEM ANALYSIS AND CLASSIFICATION

Objective

This experiment is divided into four simulations:

1. Discrete-Time System

2. Causal/Non-Causal System

3. Analysis of Discrete System

4. Analysis of Continuous-Time System

Experiment 4.1

Discrete -Time System

Program:

```
% Discrete-Time System
clc;
clear;
close all;
% Input sequence
n = 0:10;
x = [1 2 3 4 5 6 7 8 9 10 11];
% Discrete-Time System
y = filter([1 1],1,x);

% Display output
disp('Output Sequence:');
disp(y);
% Plotting
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');
ylabel('Amplitude');
```

subplot(2,1,2);

stem(n,y,'filled');

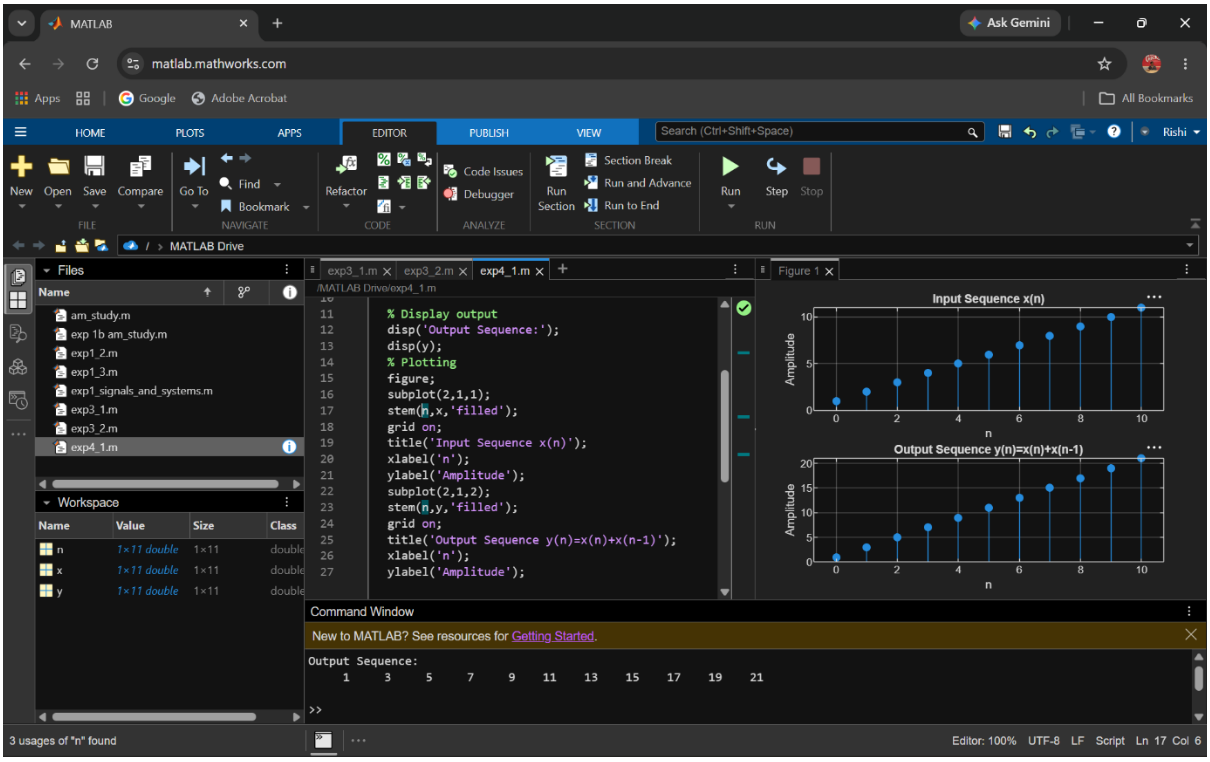
grid on;

title('Output Sequence y(n)=x(n)+x(n-1)');

xlabel('n');

ylabel('Amplitude');

Output:



Experiment 4.2

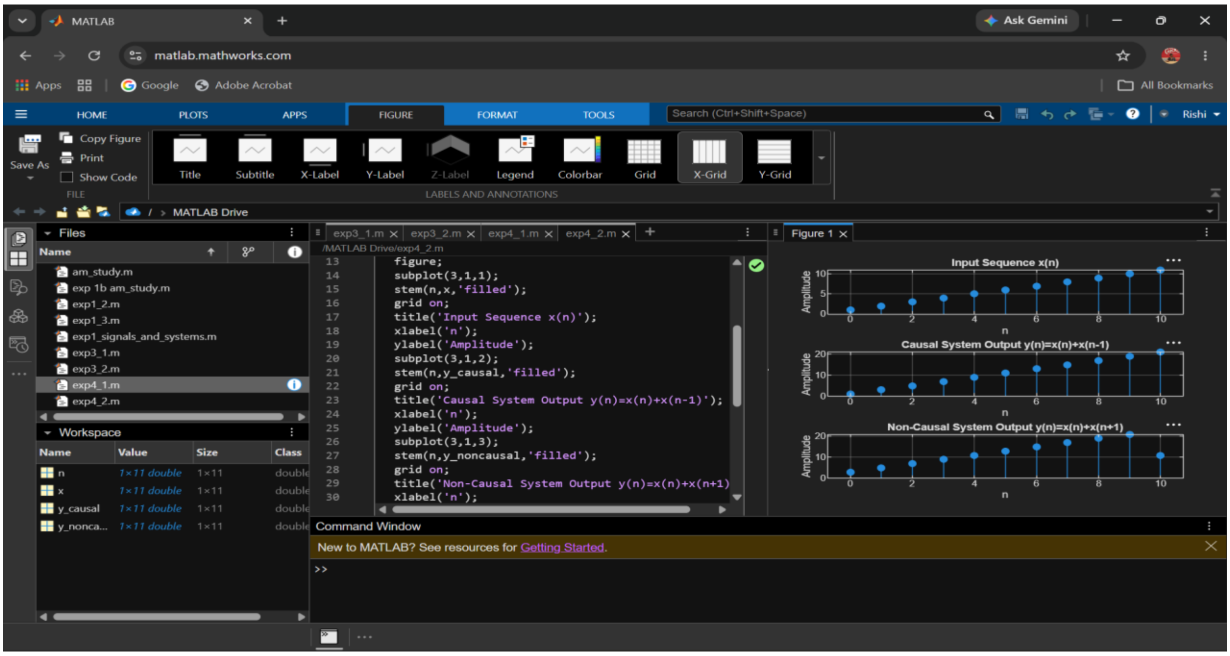
Causal/Non-Causal System

Program:

```
% Causal and Non-Causal Systems

clc;

clear;

close all;

% Input sequence

n = 0:10;

x = [1 2 3 4 5 6 7 8 9 10 11];

% Causal System: y(n)=x(n)+x(n-1)

y_causal = filter([1 1],1,x);

% Non-Causal System: y(n)=x(n)+x(n+1)

y_noncausal = x + [x(2:end) 0];

% Plotting

figure;

subplot(3,1,1);

stem(n,x,'filled');

grid on;

title('Input Sequence x(n)');

xlabel('n');

ylabel('Amplitude');
```

```
subplot(3,1,2);
stem(n,y_causal,'filled');
grid on;
title('Causal System Output y(n)=x(n)+x(n-1)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,3);
stem(n,y_noncausal,'filled');
grid on;
title('Non-Causal System Output y(n)=x(n)+x(n+1)');
xlabel('n');
ylabel('Amplitude');
```

Output:



5. System Response and LTI Properties

Objective

This experiment is divided into four simulations:

1.Impulse Response of the System

2. Step Response of the System

3. Verification of Linearity Property of LTI Systems

4. Analysis of Discrete-Time System

Experiment 5.1

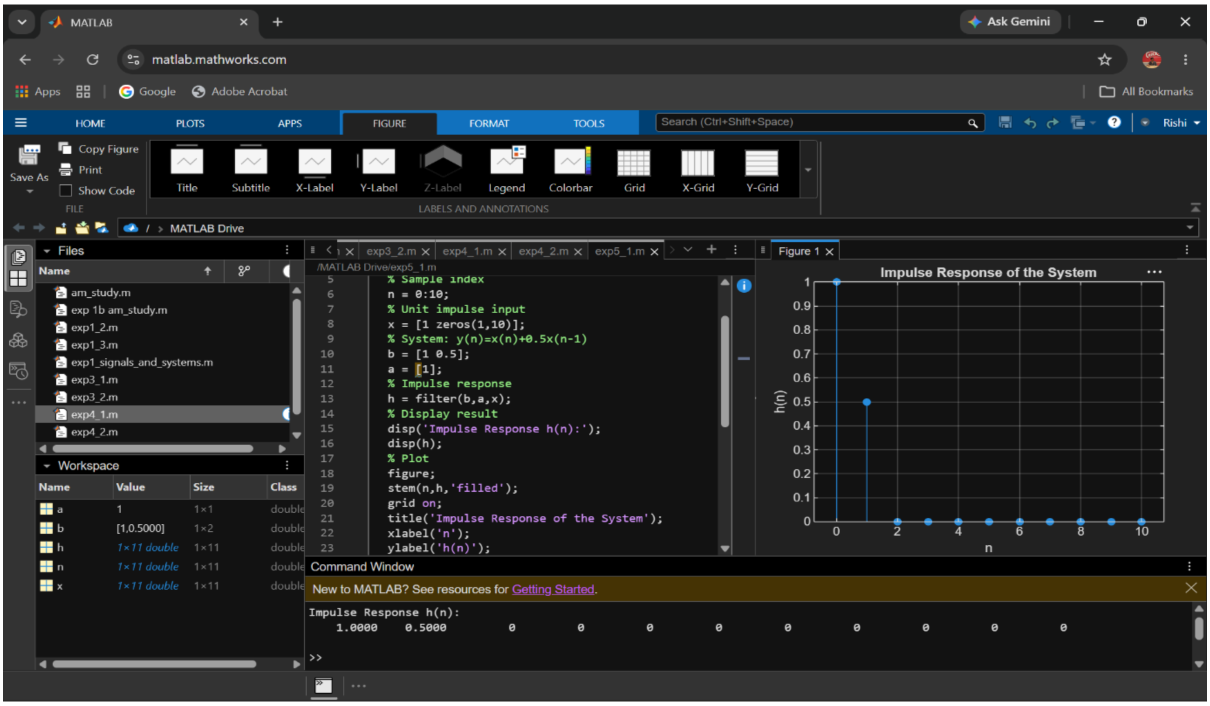
Impulse Response of the System

Program:

```
% Impulse Response of the System

clc;

clear;

close all;

% Sample index

n = 0:10;

% Unit impulse input

x = [1 zeros(1,10)];

% System: y(n)=x(n)+0.5x(n-1)

b = [1 0.5];

a = [1];
```

```
% Impulse response

h = filter(b,a,x);

% Display result

disp('Impulse Response h(n):');

disp(h);

% Plot

figure;

stem(n,h,'filled');

grid on;

title('Impulse Response of the System');

xlabel('n');

ylabel('h(n)');
```

Output:



Experiment 5.2

Step Response of the System

Program:

```
% Step Response of the System

clc;

clear;

close all;

% Sample index

n = 0:10;

% Unit step input

x = ones(1,length(n));

% System: y(n)=x(n)+0.5x(n-1)

b = [1 0.5];

a = [1];

% Step response

y = filter(b,a,x);

% Display result

disp('Step Response y(n):');

disp(y);

% Plot

figure;

subplot(2,1,1);
```

stem(n,x,'filled');

grid on;

title('Unit Step Input u(n)');

xlabel('n');

ylabel('Amplitude');

subplot(2,1,2);

stem(n,y,'filled');

grid on;

title('Step Response of the System');

xlabel('n');

ylabel('y(n)');

Output Waveform: